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SUBSECTION 4.8 PRECOMMISSIONING CHECK AND COMMISSIONING PROCEDURES

4.8.1. GENERAL

The Bidder shall be responsible for the safe and efficient setting to work of the whole of the plant and equipment. The methods adopted shall be to the approval of the Owner or his representative on Site, and shall be in accordance with any safety and permit regulations in force by the Owner on the Site.

The Bidder shall give sufficient detail in the program as required in this Contract Document to satisfy the Owner that the sequence and duration of the proposed commissioning activities are logical and realistic. Details of the Bidder's commissioning staff necessary to achieve the program shall be given.

The Bidder is required to co-operate with the Owner by organizing and setting-up the unassigned inputs of the Data Acquisition and Display System and by the production of application programs for temporary logs and display, for the periods of pre-commissioning, commissioning, and reliability runs.

At least four (4) months before commencing the commissioning of any plant or equipment, the Bidder shall submit full comprehensive procedure of pre-commissioning checks as applicable to each item of the plant and equipment provided. These procedures shall then be used during pre-commissioning as a guide to the methods to be followed and to record the actual activities carried out with the appropriate date, together with details of all work yet to be completed, variations and modifications to design conditions. The procedures should describe the prerequisites for that particular test:

- (1) Tests which should have been previously satisfactorily completed
- (2) Systems which must be live, or charged i.e. fuel tank full, live steam available
- (3) Special precautions required e.g. safety, or erection of temporary equipment e.g. for flushing
- (4) Reference to relevant drawings

Each procedure shall be accompanied by a check list. The check list shall include as a minimum the following:

- (1) Contract title
- (2) Plant item/system name
- (3) Schedule of checks to be carried out and commissioning procedure
- (4) Space for data obtained during checks to be entered and initialed
- (5) Space for date and names of persons carrying out and witnessing tests

- (6) Item/system data e.g. expected range of results
- (7) Space for remarks

The Bidder shall provide at Site the necessary certified standard instruments and gauges and any other equipment necessary for checking the installation as and when required and shall operate the plant and carry out the tests to the satisfaction of the Owner. All such test instrumentation shall be calibrated in accordance with the requirements of recognized International Standards Bureau with appropriate certification.

Each witness point activity on the procedure, when completed to the satisfaction of the Owner's Representative, shall be signed and dated by the Bidder and countersigned by the Owner's Representative.

The Bidder shall also submit for information a schedule of commissioning procedure that he proposes to follow when bringing into service groups of plant items during the commissioning period. These schedules shall detail the tests necessary to ensure the complete and satisfactory commissioning of each section of plant and shall detail all operational limitations. The schedules shall be used during commissioning which shall only commence when the relevant pre-commissioning check schedules have been completed. The procedures and check lists in the commissioning schedule shall take the same format and layout as described above for pre-commissioning.

Each witness point activity on the commissioning procedure schedule when completed to the satisfaction of the owner's Representative, shall be signed and dated by the Bidder and shall be countersigned by the Owner's Representative.

The commissioning procedure shall ensure that the commissioning of any section of the Works does not interrupt the normal commercial operation of any previously commissioned section(s).

The Bidder shall remain fully responsible for the operation, maintenance and cleaning of equipment as necessary e.g. strainers, throughout the commissioning period up to the time of takeover.

The Bidder shall prepare a coordinated 'Permit to Work' system during the course of the site works in order to establish control over the data access to all plant that has been commissioned, charged with system fluid, made live or energized, and to ensure both personnel safety and security of plant operation. This system shall be reviewed and approved by the Owner.

The Bidder shall fully comply with such safety procedures and permit systems and shall accept and accommodate the resulting limitations on access to his plant without claim on the Owner.

When the commissioning of each section of the Works is complete and before the 30 day Reliability Run is commenced, the Bidder shall carry out such preliminary tests as are necessary to establish that the plant is functioning correctly and efficiently, and shall make any adjustments required.

The Owner will provide free of charge to the Bidder only the necessary fuel and electrical power for the Bidder to set to work the main and ancillary plant and subsequent reliability and performance tests. If further preliminary runs and tests are required due to the Contract Works not complying with the Specification, the Owner may call upon the Bidder to pay for the additional charges incurred by the Owner.

The Bidder shall supply all test equipment, paper charts, continuous stationary, any magnetic tape and discs, tools, chemicals, lubricating oils and greases and other materials required for the aforementioned operations.

The Owner will make available load for the purpose of these tests but gives no guarantee that the availability will be continuous or on any specific load, however, every effort will be made to give a load as near as possible to the MCR rating of the machines.

The timing for implementation the tests shall be mutually agreed between the Bidder and the Owner.

Within two (2) months of the completion of commissioning tests, the Bidder shall submit a comprehensive report with a completed set of signed check sheets and procedures.

For purposes of this Contract, the provisions of this section will apply to plant supplied from nominated sub-Bidders.

4.8.2. BIDDER'S SITE SUPERVISORY STAFF

During the commissioning and subsequent testing of any item of plant, the Bidder shall provide the services of any special supervisory staff necessary for the purpose of ensuring proper commissioning and the satisfactory completion of all tests. The cost of such services is deemed to be part of the Bid Price for erection and commissioning of Plant.

4.8.3. COMMISSIONING TESTS OF STEAM TURBINE GENERATOR

4.8.3.1 General

The tests listed here are the minimum, and further tests specified or implied in other part of the Specification/Technical Requirement are required by the Owner.

The Bidder shall carry out other tests as implied in other parts of the Specification or which are necessary for the safe operation of the Plant.

Tests shall be applied to steam turbine generator.

4.8.3.2 Alarm and Protective Devices

(1) Low Lubricating Oil Pressure

The system shall be tested by artificially subjecting the sensing device to a reduction of lubricating oil input pressure with the turbine in operation.

(2) Low Lubricating Oil Level

The system shall be tested by artificially subjecting the sensing device to a reduction of oil level with the turbine in operation.

(3) High Temperature

This test applies to the lubricating oil system, the bearing metal, turbine casing and generator. Each sensing head shall be removed and heat applied to it externally until the set temperature is reached.

(4) Emergency Overspeed Trip

No setting of the emergency overspeed trip shall be permitted during the test. Precautions shall be taken to ensure that the turbine will not exceed the maximum safe speed stated by the manufacturer.

The test shall be carried out at no load and by adjusting the governor setting or the leakage to bring the turbine slowly to sufficient speed. The speed shall be caused to be increased at no higher than one (1) percent per five-seconds. The speed shall be measured with a maximum error not exceeding $\pm 0.25\%$.

At reaching trip limit the gas turbine protection system shall trip the gas turbine

(5) Loss of Circulating Water

One (1) circulating water pump shall be tripped manually with the turbine operating at full load to demonstrate recovery of the plant without a trip.

4.8.3.3 Vibration

Test method shall be to the approval of the Owner with particular regard to the following points:

- (1) Location of measurement, vibration or bearing casing(s) or displacement of shafts (s).
- (2) Use of pick-up measuring acceleration or displacement speed of displacement itself.

4.8.3.4 Handling Characteristics

(1) Starting Sequence

Determination of the starting sequence shall be carried out and recorded in a test log.

The Bidder and the Owner shall reach agreement on the method of indicating significant events in the starting sequence.

(2) Starting Reliability

The Bidder shall be required to demonstrate that the automatic start-up and run-up procedure is capable of five (5), entirely trouble-free starts, one of which shall be cold starts. The hot start test shall be carried out immediately upon tripping from full load.

(3) Remote Starting

The Bidder shall demonstrate that the steam turbine generator shall be capable of being started by remote control at the Central Control Room under manual and automatic mode.

(4) Normal Load Pick-up

The capability and reliability of load pick-up shall be proved with a minimum of three (3) tests, of which at least one (1) shall be made from cold. These tests may be performed as part of any successful start in the starting reliability tests. These tests shall be performed in accordance with the operating instructions.

(5) Ramp Load Cycling

The capability and reliability of the turbine in conjunction with the HRSG under ramp load cycle as described in the Technical Requirements shall be proved with a minimum of three (3) tests made with the turbine in hot condition. These tests may be performed as part of the reliability run and carried out in accordance with the operating instructions.

(6) Load Rejection

The test shall be accomplished with the turbine working at specified load with all steam admission valves open as required and under speed governor control, by removing the load instantaneously. The test shall be carried out at 50% and 100% MCR load. All protection system and control system settings shall be at their normal values for both steam turbine and HRSG. The successful operation of the bypass system shall be demonstrated during this test. Safety valves shall not blow off during the test.

4.8.3.5 Partial Condenser Operation

The capacity and stability of operation of the steam turbine with only 50% of the condenser available shall be demonstrated to the satisfaction of the Owner.

4.8.3.6 Generator Tests

(1) Generator Temperature Rise

The generator temperature rise shall be determined by running the machine for at least six (6) hours at full (MCR) load, and for such an additional period as may be required for the plant to attain a steady temperature.

(2) AVR Response Test

A test shall be performed to determine the AVR response characteristics.

(3) One Air Cooler Out

The generator shall be run with one (1) section of the air coolers out of operation to measure its reduced output and increased temperature rise. If the hydrogen cooling is applied, the same procedure as the air coolers shall be performed.

4.8.3.7 Shutdown

The shutdown systems, including DC jacking oil systems, shall be tested by AC power failure.

4.8.3.8 Governor Tests

No adjustment of the main governor or the overspeed governor or any of their main connections shall be permitted during any governor or overspeed test or series of tests, except in the determination of speed range adjustment. Tests to determine permanent speed variation may be either increasing or decreasing load.

Immediately prior to the tests, the thermodynamic conditions shall be controlled and thereafter adjusted and eventually restored to normal. The tachometer or revolution counter and the whole governing system shall be examined and placed in a reliable working condition.

(1) Speed Governor Droop

The droop shall be measured and checked to be within the specified limits stated in the Technical Requirements.

(2) Constant Speed Regulation

During test, the only governor adjustment shall be that initially required for establishing the rated speed at corresponding rated power outputs.

Tests shall be carried out to determine the following:

- Maximum and minimum, manually or remotely adjustable sustained governed speed at no load
- Permanent and momentary speed variation
- Dead band

(3) Variable Speed Regulation

Tests shall be carried out to determine the following:

- General stability of the governed speed control at given settings, including idling settings, within the region of operation as given in (2) below. The settings are to be agreed between the Bidder and the Owner.
- Maximum and minimum, manually or remotely adjustable sustained governed speed at the appropriate load.
- Maximum momentary increase in the governed speed above the rated speed when the steam control setting or steam regulating valve is suddenly shifted from idling to the rated speed position under conditions of no load.
- Maximum momentary decrease in governed speed below the idling no-load speed when the steam regulating valve is suddenly shifted from the rated speed position at steady rated power output to the idling no-load setting.
- Dead band.

4.8.3.9 Steam Strainers

Temporary steam strainers shall be supplied and fitted to the steam circuits to prevent any damage to the turbine blading during the initial commissioning period. Temporary strainers shall also be supplied and fitted in the auxiliary steam supply (e.g. steam to air ejectors and gland steam system).

The temporary strainers shall have openings smaller than those of the permanent strainers, and shall be suitable for passing sufficient steam to generate the full rated output without exceeding a differential pressure which is liable to cause failure of the strainers within a period of at least 1500 hours of operation at that differential pressure.

The Bidder shall be responsible for cleaning all steam strainers during the initial commissioning period and the subsequent removal of the temporary strainers.

4.8.4. COMMISSIONING TESTS FOR HEAT RECOVERY STEAM GENERATOR (HRSG)

In addition to tests specified or implied in other parts of the Specification, the following tests shall performed on HRSG to the satisfaction of the Owner:

- (1) Start-up sequence of HRSG and auxiliaries on automatic control including control damper change over from bypass stack to HRSG inlet.
- (2) Shutdown sequence of HRSG and auxiliaries on automatic control including control damper changeover from HRSG inlet to by-pass stack.
- (3) Automatic protection of HRSG against running dry.
- (4) Operation of safety valves.
- (5) Operation of each steam bypass system.

4.8.5. COMMISSIONING TESTS FOR COMBINED CYCLE BLOCK

In addition to tests specified or implied in other parts of the Specification, the following tests shall be performed on combined cycle power plant to the satisfaction of the Owner:

- (1) Start-up sequence of the block on automatic control up to full load. The method of timing and indicating significant events shall be agreed between the Owner and the Bidder.
- (2) Ramp load pick-up as described in the Technical Requirements. These shall be proved by a minimum of two (2) and maximum of five (5) tests of which at least one (1) must be made from cold condition. The tests shall be performed in accordance with the approved procedures.
- (3) Shutdown sequence of the block on automatic control and loss of AC power supply. The method of timing and indicating significant events shall be agreed between the Owner and the Bidder.
- (4) Load rejection from 50% and 100% MCR. The test shall be performed with the plant in hot condition. The resultant maximum transient system frequency and the ultimate steady frequency shall be recorded. During the test, no adjustment shall be made to the main governor or the overspeed governor or any of their connections.
- (5) Operation of the gas and steam bypass systems, including a steam turbine trip.
- (6) Maximum primary response in modified variable pressure control mode to support stabilization of electrical grid frequency in line with the requirements of this Bid Document.

(Last page of Subsection 4.8)